

PHASE 303: THE LUNAR BUGGY — THE GENERAL

the machine that out-charges the dust, renews its own wheels, and banks the dark

Rovers die of dust, cold, and darkness long before they die of mileage. Curiosity's wheels tore open on aluminium skins; Perseverance flew a thicker redesign as the confession. NASA's best dust answer is a powered electrode curtain licensed for fixed panels, and its surface-mobility shortfall — 1304, robust, high-progress-rate, long-distance, autonomous — sat at #9 on the agency's own integrated list because no one had the whole machine. This chapter seats the whole machine: one buggy, four seated legs, a recovery kit, and its own pop-up eyes.

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1 • The Law

A surface machine that engineers its own potential, renews its own tread, and banks its own night dies of nothing the surface can bring. The buggy does not fight the dust — it out-charges it. It does not outlast the wheel — the wheel renews itself. It does not survive the night — the forecourt banks it.

2 • The Name — the General

The buggy is named The General — for General Antonio Luna, one of the greatest Filipinos in history, who defied his surrendering government and fought the Americans, and was assassinated by his own government for it (Cabanatuan, 5 June 1899). The controversy is deliberate, and controversy is good marketing: the echo of the General Lee gets the machine talked about. But the name itself is an honor and a standard — the General does not surrender.

3 • The Chassis — the Fish

The chassis is the Fish family (Phase 5): a rigid chassis ring with a riser-rod canopy, an isolated cabin volume in the head — the driver's cabin when crewed (Phase 5 doctrine: the head contains the cabin for the occupant/driver) — and a tail compartment for power and the collector payload. And at the heart of the ring sits the component that makes a Fish a Fish: the axle-less spherical inertial shift drive — the

gimbal sphere held on riser pins, the inertial-shift gyros within, with the dead-man safety gyro (Phase 5). In one-sixth g the sphere does the work gravity will not: it generates the buggy's own directional mass and presses the machine down onto the regolith — a bare rover floats and skates, the Fish holds the ground. And because the drive rides inside the ring, the hull never fights it: the body can rotate full circle around the sphere without taking gyro torque shear through the wheels — the rover-versus-fish law, seated. When no crew rides, the driver is GK3/Grained AI: no body needed. Navigation rides the corridor-survey maps (Phase 211), mapped to metre scale before the first pod lands, with graze-roll terminal guidance for the last metres.

4 • The Wheels Are the Ground Brush

The wheels are the Phase 2 Self-Treading Endless Wheel, space/rover variant: metal frame, metal rods, hexagonal tubes from tread face to hub with spring-fed taper-locked rods inside. Every rod is a grounding contact pressed to the regolith — and this is the wheel's second job: the buggy's electrical connection to the ground is self-renewing hardware, not a brush that wears out. The 5-degree taper lock means the contact cannot fall out, only wear out, and the feed renews it as it wears. Rods and springs are silicone-boot jacketed — the CV-boot precedent: grease in, dust out — with graphite grease per the lubricant law (Tru-Blu nevermelt bentonite class, matched to the medium). Same wheel class as the Mars rover wheel gives the seated 50-year estimate against its ~15. Long-distance dies of nothing.

5 • We Are the Curtain

The whole buggy holds one half of the charge (Phase 208). A vehicle in a charged environment is one plate of a capacitor whether it likes it or not — so the plate's potential is engineered, never accidental. Held like-polarity charge repels the dust outright: no sweeping, zero watts. When a second field is wanted, small high-speed laser arrays sweep the ground ahead and set the ground charge on the machine's terms; the rods close the loop to ground. The existence proof is seventy years old: an unpowered fluorescent tube glows under a high-voltage line — a floating object in a field is worked upon by the field, and the field is readable for free.

6 • The Bottle Frame

The frame is 100 mm tube, pressure class, filled with backup oxygen — the structure is the breathing-gas reserve. Law: the tubes carry per-segment isolation valves, because a puncture must never be allowed to be a structural event and a life-support event at the same time. Class estimate: about 7.9 litres per metre of tube; at 200 bar roughly 2.1 kg of oxygen per metre of frame. This reserve is not counted in the trip — it is the cans of food in the storm cellar, a gift by design, not a requirement.

7 • Cold and Darkness — the Forecourt

The buggy's home plate is the rover forecourt: the charging stand (Phase 174), Night Bank power through the 14-day dark, and the freeze answer and parking law (Phase 261) governing the machine when the sun dies. The buggy never winters alone; it parks where the bank is.

8 • The Recovery Pair

Two winches, not one: a normal winch for normal pulls, and the winch-rocket — a small rocket that fires a grappling hook to lay the line where no machine can drive. Vacuum and one-sixth g make the rocket line-layer easy; the marine line-throwing rocket is a century old and acknowledged as heritage — the lunar recovery application is the new seat.

9 • The Rescue Fit-Out

The General is the seated surface rescue vehicle (Phase 5 amendment), and the fit-out is law, not option: reflection, solar and batteries — a self-sufficient field power plant; prill-tech cooling, solid-state, no cryogenics; oxygen, water and food for days-class survival aboard. The medical stretcher pod lifts straight off the ship's wall and bolts to the buggy's side — the seated golden-hour trauma pod, so the rescue carries the emergency room, not a first-aid kit. The pod's form class: the clear-domed casualty transport capsule — the infant-transport BabyPod lineage, acknowledged heritage — a transparent-lidded stretcher pod on a tubular side rack, bolted between gimbal and tail. The pod rides warm: electrotherm heating panels — the sandwich-layer thin-film carbon elements (Michael Bell, Electrotherm Pty Ltd, mid-1980s, aircraft de-icing ancestry, unpatented, acknowledged in perpetuity; Phases 61 and 91 are the file's descendants) — because on the moon a casualty dies of cold before anything else, so the stretcher is heated, not just carried. And the crown: with G. The gimbal architecture permits the cabin ring to spin — the buggy doubles as a centrifuge, and earth blood treatments (settling, separation, centrifuge-class blood processing) run at Earth-normal g in a one-sixth-g world.

10 • The Eyes — Phase 302 Tap Drones

Every excursion carries poppers, and that is mandatory law: lose radio, lose line-of-sight, lose your way — pop one. A quick burst up, then small taps to keep it up, like tapping a balloon; the drone auto-taps while the camera looks, and carries strobe lights for long-way-out recovery. The long-distance class scouts far ground in ballistic hops. The gas is CO₂, a byproduct — the eyes are free. The point of the law is oxygen: you only have so much, and you do not spend it searching.

11 • Operations

Every sortie: poppers aboard before wheels turn — no exceptions. Dust operations are passive-by-design: the held charge is the default state, not a mode. The laser ground field runs when the dual field is wanted. Tread renewal is the only wear service; there is no tyre to change. The buggy parks at the forecourt by the parking law, charges from the stand, and breathes from its own frame only when it must. Rescue sortie: pod aboard, panels warm before the casualty is.

12 • Materials

Metal frame and rods (rover variant, Phase 2). Silicone boots on rods and springs. Graphite grease per the lubricant law — Tru-Blu nevermelt bentonite class, matched to the medium, never one universal name. Chassis and wheel printed from Phase 274 fifty-year feedstock blocks. Laser array heads off the shelf. 100 mm pressure-class tube for the bottle frame. Gimbal sphere, riser pins and gyro cluster per Phase 5. Ship-wall stretcher pod and electrotherm thin-film panels for the rescue fit.

Owed to the Bench

Polarity management across the terminator — dayside dust charges positive, nightside negative, and the held charge must read and oppose. Ground return through bone-dry dielectric regolith — the laser ground field is the answer; the bench must prove it. Wear-rate numbers behind the 50-year estimate. Thermal-cycle rating of the Fish canopy. Tap-rate and bottle sizing for the poppers. Isolation valve classes for the bottle frame.

THE SCALP: the buggy doesn't fight the dust — it out-charges it; it doesn't outlast the wheel — the wheel renews itself; it doesn't survive the night — the forecourt banks it; and nobody spends oxygen searching — the poppers see for you. Its name is a promise — the General does not surrender.